

cross_residual	skew_residual
8.881784197001252323e-16	1.041851292966825522e-15
1.047382306668854030e-15	1.016044584868286263e-15
4.335559509131366993e-16	6.004449063730082196e-16
1.332267629550187849e-15	4.102795222832630006e-16
4.577566798522237452e-16	7.301351988545593167e-16

The simulation gives residuals of approximately 10^{-15} for both identities, which is very close to machine precision. This strongly supports that the identities are satisfied by the rotation matrices generated by MuJoCo.

The simulation only tests the identities for a finite number of numerical examples. A small but nonzero residual does not mean that the identity is false. This error is expected because computers use finite-precision floating-point arithmetic, which introduces small rounding and numerical errors. Therefore, a residual close to zero supports the identity numerically, while the mathematical proof is what establishes that it is true in general.